

Assignment 1: Opinion Network Formation

Dynamical Processes and Complex Networks (DPCN)

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GitHub Repository: <https://github.com/garimamittal13/Dynamical-Processes-and-Complex-Networks>

Abstract

This report details the construction and analysis of an opinion network derived from a survey capturing perspectives on Technology, Education, Ethics, and Environment. By representing the 60 individual survey questions as nodes and their Pearson correlations as edges, we mapped the cognitive landscape of the respondents. The raw categorical Likert-scale data was cleaned, quantified, and processed to generate a correlation matrix. Using an absolute correlation threshold of $r > 0.33$, we constructed an undirected graph consisting of 60 nodes and 450 edges. Topological analysis utilizing Degree, Betweenness, and Eigenvector centralities revealed that opinions on Ethics and Environment exhibit the highest internal consensus and act as predictive bridges across other domains. This methodology successfully highlights the underlying structures in class opinions, satisfying the assignment's analytical objectives.

1 Introduction

Understanding the ideological alignment of a student body is crucial in an era defined by rapid technological advancement and complex ethical dilemmas. This assignment seeks to uncover the latent relationships between differing opinions by employing complex network analysis. Rather than evaluating individual survey responses in isolation, this study constructs a macro-level network where the interconnectedness of opinions can be mathematically quantified and visually explored. The objective is to determine whether distinct ideological communities exist within the class and to identify which specific beliefs act as foundational pillars for broader worldviews.

2 Dataset Documentation

2.1 Overview

This section describes the interpretation and preparation of the survey data provided in the assignment. The dataset encompasses class opinions across four primary domains, designed to capture a holistic view of the respondents' socio-technical perspectives:

- **Technology (Prefix: T):** Focusing on AI adoption, automation, and data privacy.
- **Education (Prefix: E):** Evaluating pedagogical preferences, such as project-based vs. rote learning.
- **Ethics (Prefix: S):** Assessing views on inclusivity, societal responsibility, and technological accountability.
- **Environment (Prefix: V):** Measuring attitudes toward sustainability and climate action.

2.2 Data Cleaning and Preprocessing

The original dataset (`Survey_Results_UC.csv`) contained 96 responses. Data cleaning involved the following systematic steps:

- **ID Removal:** The `id. Response ID` column was dropped to anonymize responses and focus purely on opinion variables.
- **Categorical Mapping:** Likert-scale string responses were converted into a quantitative spectrum to enable mathematical correlation. Specifically:

Strongly Disagree (1), Disagree (2), Neutral/No Comments (3), Agree (4), and Strongly Agree (5).

- **Missing Data Handling:** Rows consisting entirely of empty responses were dropped from the dataset. Any remaining isolated missing values were imputed with 3 (Neutral) to prevent localized data loss during correlation matrix calculation.

2.3 Feature Selection and Justification

All 60 survey questions were selected as individual features (nodes). Rather than aggregating scores by category (which would artificially compress the data into four monolithic nodes), treating each question as a distinct node allows for granular observation. For instance, this approach enables us to observe if a specific view on autonomous vehicles correlates directly with a distinct ethical stance on inclusivity, revealing interdisciplinary ideological bridges.

3 Pipeline Followed

3.1 Network Definition

In our undirected network, **nodes** represent the 60 individual survey questions. An **edge** exists between two nodes if the absolute Pearson correlation coefficient between the students' responses to those two questions exceeds a predefined threshold. This ensures that only statistically significant similarities (or strong inverse relationships) in opinions are mapped.

3.2 Construction Process

The raw data was transformed into a graph structure via the following pipeline:

1. **Step 1 - Correlation Matrix Generation:** We computed a 60×60 Pearson correlation matrix using the quantified student responses. Pearson was selected over Spearman to capture the linear magnitude of agreement across the uniform 1-to-5 scale.
2. **Step 2 - Threshold Optimization:** To prevent a densely saturated "hair-ball" graph, we applied an absolute correlation threshold of > 0.33 . Iterative testing revealed that thresholds above 0.45 caused severe network fragmentation, while thresholds below 0.30 resulted in uninterpretable density. The 0.33 mark represents the optimal balance of topological structure and legibility.

3. **Step 3 - Graph Initialization and Centrality Calculation:** We populated an undirected graph using the thresholded matrix. We then calculated Degree Centrality (to find predictor questions), Betweenness Centrality (to find bridging questions), and Eigenvector Centrality (to find highly influential opinions).

3.3 Algorithm and Tools Used

We utilized Python frameworks to optimize the layout algorithm. Key libraries included:

- **Pandas:** For data ingestion, string manipulation, and correlation matrix math.
- **NetworkX:** For graph construction, topological algorithms, and metric calculations.
- **Matplotlib:** For advanced layout spacing. We deliberately restricted the iterations of the `spring_layout` algorithm and implemented mathematical coordinate normalization to force the nodes apart, ensuring the dense central core remained visually legible.

4 Analysis and Visualizations

4.1 Graphical Representations

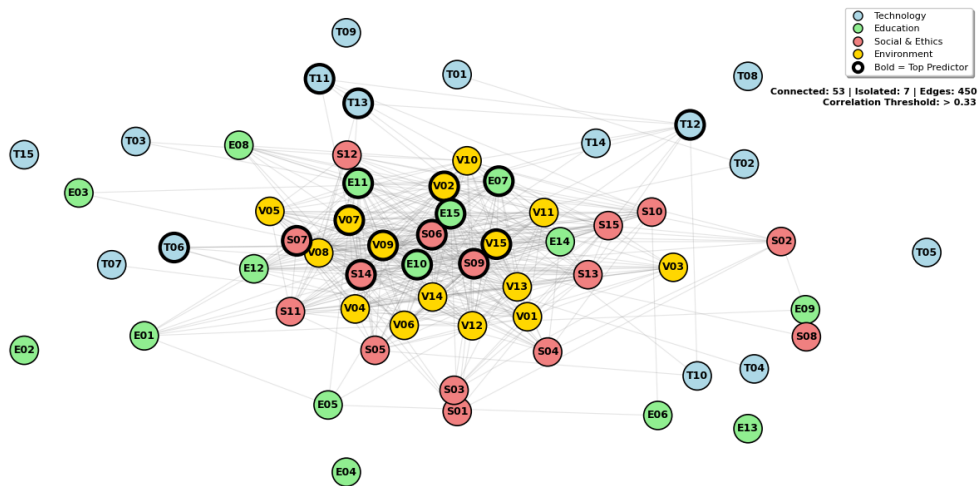


Figure 1: Opinion Network constructed with a Pearson correlation threshold of $r > 0.33$. Nodes are color-coded by category, with thick bold borders highlighting the top predictor nodes for each domain.

4.2 Key Network Metrics

Our network topological analysis yields the following global metrics at the 0.33 correlation threshold:

- **Number of Nodes (N):** 60
- **Number of Edges (E):** 450
- **Average Degree ($\langle k \rangle$):** 15.0 (calculated as $2E/N$)
- **Connected Components:** 1 main giant component (53 nodes) and 7 isolated nodes.
- **Network Density:** 0.254 (indicating that approximately 25% of all possible strong correlations exist within the respondent pool).

Top Nodes by Category: The table below highlights the most central nodes computationally identified from each domain in our network.

Domain	Top Predictors (Degree)	Top Bridges (Betweenness)	Top Influencers (Eigenvector)
Technology	T12, T06, T11, T13	T02, T12, T11, T06	T12, T06, T13, T11
Education	E10, E11, E15, E07	E10, E12, E11, E01	E10, E15, E11, E07
Ethics	S09, S14, S06, S07	S09, S10, S14, S05	S09, S14, S07, S06
Environment	V09, V07, V15, V02	V01, V07, V02, V10	V09, V14, V13, V07

5 Results and Discussion

5.1 Global Structural Findings

The overarching structural pattern of the network reveals a dense, highly interconnected core dominated by Ethics (S) and Environment (V) nodes, surrounded by a sparser periphery of Technology (T) and Education (E) nodes. The high density within the S and V clusters mathematically demonstrates that respondents who agree on one ethical or environmental issue are highly likely to agree on the others.

5.2 Domain-Specific Ideological Analysis

Ethics and Environment (The Core): The dense clustering in these sectors suggests the class shares strong, unified ideological frameworks. Nodes **S09** ("Universities should promote an inclusive environment") and **V09** ("Products should be designed to encourage reuse") possess the highest degree centralities in their respective domains. Consequently, these serve as primary "Predictor Nodes"—knowing a student's stance on S09 allows for highly accurate statistical predictions of their stances on dozens of other survey questions.

Technology (The Bridges): While Technology opinions were more fragmented, the Betweenness metrics identify questions like **T02** (Generative AI as learning aids) and **T12** (Stricter AI regulations) as vital bridges. These nodes act as ideological translators, connecting the highly technical nodes to the core ethical cluster of the class.

Education (The Unifier): In the Education sector, **E10** (Prioritizing innovation over rote learning) stands out across all three centrality metrics (Degree, Betweenness, and Eigenvector). This indicates that moving away from rote learning is a core, unifying sentiment for the class that correlates heavily with pro-innovation stances in the Technology domain.

5.3 Analysis of Isolated Opinions

Notably, 7 questions were completely isolated at the 0.33 threshold: **T05, T08, T09, T15, E02, E04, and E13**. These nodes exclusively belong to the Technology and Education sectors. The isolation of these nodes indicates that opinions on these highly specific topics operate entirely independently of the respondents' broader worldview; an individual's stance on climate change or ethical inclusivity offers no predictive power regarding their stance on these 7 outlier issues.

5.4 Methodological Limitations

Our primary limitation is the inherent sensitivity of the correlation threshold. Furthermore, treating ordinal Likert data as continuous numeric variables introduces standard mathematical biases, as the psychological distance between "Neutral" and "Agree" may not exactly equal the distance between "Agree" and "Strongly Agree."

6 Conclusion

This network analysis successfully mapped the latent ideological structures of the surveyed class. The findings highlight that while opinions on specific technological implementations remain varied and sometimes isolated, the student body is anchored by a dense, highly correlated consensus on ethical responsibility and environmental sustainability. Future iterations of this study could employ community detection algorithms, such as Louvain or Girvan-Newman, to dynamically cluster respondents rather than the questions themselves, further illuminating demographic divides within the dataset.

7 Individual Contribution

The specific tasks completed by each group member are detailed below:

- **Dataarnoor Singh Oberoi:** Led the data engineering pipeline and structural network definition. Handled categorical-to-numeric mapping, generated the Pearson correlation matrix, and iteratively tested the correlation thresholds ($r > 0.33$) to optimize network density. Drafted the Dataset Documentation and Methodology sections.
- **Garima Mittal:** Led the visual-spatial network analysis and structural layout. Decoupled the layout algorithms in Matplotlib to accurately represent ideological distances, analyzed the significance of the isolated nodes, managed the GitHub repository, and finalized the formal LaTeX typesetting.
- **Saarthak Sabharwal:** Led the topological metric analysis using NetworkX. Computed all node centralities (Degree, Betweenness, Eigenvector) to identify ideological predictors and bridges. Synthesized the mathematical outputs into the core ideological findings and drafted the Results and Discussion section.